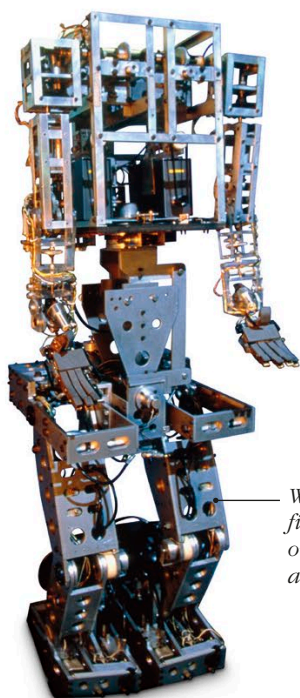




WABOT-1

In 1972, scientists at Waseda University, Japan, unveiled Wabot-1, the world's first full-scale humanoid intelligent robot. It was the first android—a robot that resembled humans in some ways. It was able to walk, communicate with a person using its artificial mouth, measure distances and directions to objects, and grip and transport objects with its hands.

Wabot-1 was the first robot to walk on two legs like a human.



MILITARY ROBOTS

In the US, Boston-based Denning designed the Sentry robot (right), in 1985, as a security guard that can patrol for up to 14 hours and radio in alerts if it senses anything out of the ordinary. Today, South Korea uses Samsung armed military robots to guard the border with North Korea.



IBM's Watson robot competing against human contestants

ROBOTS VS. HUMANS

In 1997, IBM's Deep Blue robot beat the world chess champion Garry Kasparov in a match. In 2008, IBM's Watson robot defeated human panelists in the TV quiz show *Jeopardy!* Machines have also beaten humans at board games Scrabble and Go.

LIFELIKE ROBOTS

Some of the latest robots are able to talk like humans, walk like humans, and express a wide range of emotions. In 2016, the Chinese presented Jia Jia, an AI (artificial intelligence) humanoid (right), which claimed to be the most humanlike robot ever made.

FAST FACTS

- Robots are not a new idea—the Italian Leonardo da Vinci sketched “mechanical knights” in 1495.
- Japan leads the world in robotics—much of the research is led by government agencies, particularly the Trade Ministry.
- Research is ongoing into nanobots—robots so tiny that they could work inside a human body to keep it healthy.

